

Vacuum

Predicting Sputtering Yield and Penetration Depth in 316L Stainless Steel under Low-Energy Nitrogen Ion Bombardment: A Molecular Dynamics-Informed Machine Learning Approach

--Manuscript Draft--

Manuscript Number:	VAC-D-26-02042
Article Type:	Full Length Article
Keywords:	Molecular dynamics; Machine learning; Sputtering yield; Penetration depth; Nitrogen ion bombardment; 316L stainless steel
Abstract:	Low-energy nitrogen ion bombardment is essential in plasma surface engineering, but predicting sputtering yield and penetration depth remains challenging. In this proof-of-concept study, we develop a molecular dynamics (MD)-informed machine learning (ML) framework to predict these responses in 316L stainless steel. MD simulations across four energies (50–300 eV) and six impact angles (0°–75°) generated a dataset for training Linear, Decision Tree, and Random Forest regressions. Linear Regression gave the best sputtering yield prediction ($R^2 = 0.586$), while Random Forest performed best for penetration depth ($R^2 = 0.747$, RMSE = 2.196 Å). Although dataset size limits predictive accuracy, this work demonstrates the feasibility of MD–ML integration as a computationally efficient surrogate modeling approach for rapid screening of ion irradiation conditions.

Predicting Sputtering Yield and Penetration Depth in 316L Stainless Steel under Low-Energy Nitrogen Ion Bombardment: A Molecular Dynamics-Informed Machine Learning Approach

J. Taghinejad ^{a,b*}

^a Department of Physics, CT.C, Islamic Azad University, Tehran, Iran

^b Department of Electromagnetic and Optical Technologies, CT.C, Islamic Azad University, Tehran, Iran

Corresponding Author. Email: javad_taghinejad@iau.ac.ir

Abstract

Low-energy nitrogen ion bombardment is essential in plasma surface engineering, but predicting sputtering yield and penetration depth remains challenging. In this proof-of-concept study, we develop a molecular dynamics (MD)-informed machine learning (ML) framework to predict these responses in 316L stainless steel. MD simulations across **four** energies (50–300 eV) and **six** impact angles (0°–75°) generated a dataset for training Linear, Decision Tree, and Random Forest regressions. Linear Regression gave the best sputtering yield prediction ($R^2 = 0.586$), while Random Forest performed best for penetration depth ($R^2 = 0.747$, RMSE = 2.196 Å). Although dataset size limits predictive accuracy, this work demonstrates the feasibility of MD–ML integration as a computationally efficient surrogate modeling approach for rapid screening of ion irradiation conditions.

Keywords: Molecular dynamics; Machine learning; Sputtering yield; Penetration depth; Nitrogen ion bombardment; 316L stainless steel

1. Introduction

Low-energy ion irradiation is a cornerstone technique in plasma surface modification, ion implantation, thin-film deposition, and ion-beam-assisted surface engineering due to its capability to precisely tailor near-surface material properties [1,2,6,33]. During ion bombardment, energetic ions transfer their kinetic energy and momentum to surface atoms, initiating collision cascades and inducing atomic-scale modifications [8–10]. However, the inherent stochasticity of ion–solid interactions complicate accurate prediction of irradiation outcomes, particularly when exploring a wide range of processing conditions. Among the key irradiation responses, sputtering yield and ion penetration depth are of primary importance. The sputtering yield—defined as the number of surface atoms ejected per incident ion—quantifies surface erosion, while the penetration depth characterizes the subsurface transport and final resting position of implanted ions [2,7–9].

Together, these two parameters provide complementary insights into energy partitioning and material modification during ion–surface interactions [1,6,8].

Austenitic stainless steel 316L is extensively utilized in industrial, biomedical, and nuclear applications owing to its excellent corrosion resistance, mechanical strength, and structural stability [17,18]. Nitrogen ion implantation has been widely investigated as a means to enhance the surface hardness, wear resistance, and corrosion behavior of stainless steels [11–13,19]. Nevertheless, energetic ion bombardment can significantly alter surface morphology and near-surface atomic structure, thereby affecting long-term performance [2,10,20]. Consequently, understanding the dependence of sputtering behavior and ion penetration depth on irradiation parameters is essential for the controlled application of nitrogen ion bombardment in surface engineering [1,7,21,35,36].

Molecular dynamics (MD) simulation is a powerful atomistic tool for investigating ion–solid interactions, providing detailed information on atomic trajectories, collision cascades, energy transfer, and surface modification mechanisms [8–10,22,34]. However, MD simulations are computationally expensive, particularly when systematic exploration of multidimensional parameter spaces is required [23,24]. In our previous work, we systematically investigated the effects of low-energy nitrogen ion bombardment on FCC 316L stainless steel as a function of incident energy and impact angle, revealing the underlying atomic-scale interaction mechanisms [34–36]. Building upon those results, the present study develops an MD-informed machine learning framework to rapidly predict sputtering yield and penetration depth without performing additional atomistic simulations [8–10].

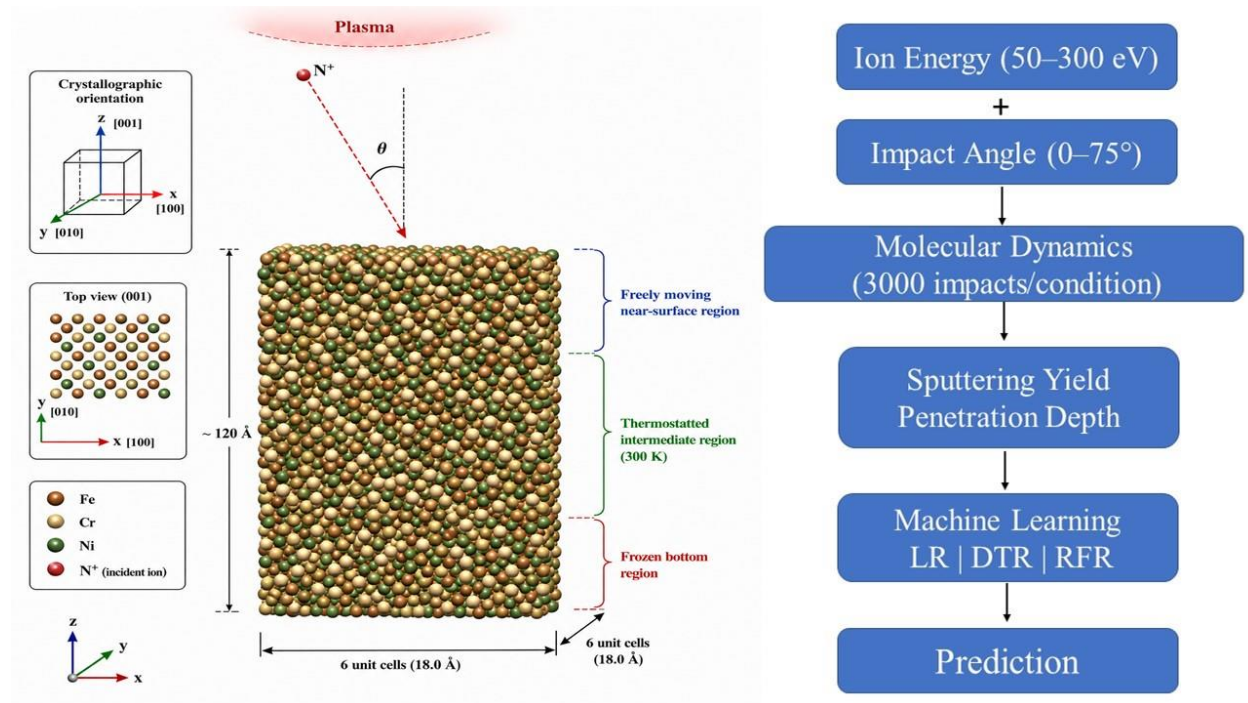
To overcome the computational limitations of atomistic simulations, machine learning (ML) has emerged as a powerful approach for developing surrogate models that establish predictive relationships between processing parameters and material responses [3,4,14,15]. ML-based regression models have been successfully applied to predict a variety of materials properties and to reduce the computational cost of simulation-driven investigations [3,4,25,26]. Recently, ML approaches have also been explored for predicting sputtering behavior using physically meaningful descriptors [5,16,27]; however, many existing studies rely primarily on experimental datasets or conventional materials descriptors, and MD-informed prediction of ion-induced surface modification remains relatively underexplored [5,16]. Moreover, sputtering yield and penetration depth are governed by distinct atomic-scale mechanisms—surface atom ejection versus subsurface ion transport—making their simultaneous prediction a nontrivial task that has not yet been systematically addressed using MD-generated datasets [28,29].

In the present study, we develop an MD-informed machine learning framework using data generated from low-energy nitrogen ion bombardment of FCC 316L stainless steel. Incident ion energy and impact angle are employed as input descriptors, while sputtering yield and penetration depth are selected as target responses. Unlike previous studies, the proposed framework is trained exclusively on MD-generated data, enabling efficient prediction of irradiation responses directly from controllable process parameters [8–10,16,34]. To evaluate the framework, three regression models with increasing complexity—Linear Regression, Decision Tree Regression, and Random Forest Regression—are systematically compared using standard statistical performance metrics [30,31]. This study demonstrates that integrating MD simulations with machine learning enables

reliable, computationally efficient, and physically interpretable prediction of sputtering yield and penetration depth under low-energy nitrogen ion irradiation, with potential applications in plasma surface engineering and ion implantation process optimization [3,5,16,32].

2. Molecular Dynamics Simulation and Dataset Preparation

- Figure 1 summarizes the overall workflow of the present study, which integrates molecular dynamics (MD) simulations with machine learning (ML)-based prediction. In this framework, MD simulations serve as an atomistic data generation tool, providing physically meaningful irradiation responses, including sputtering yield and penetration depth, under different nitrogen ion energies and impact angles. The simulation dataset, generated following the methodology established in our previous molecular dynamic's studies [34–36], is subsequently used to train regression models for rapid prediction of ion-induced surface modification responses without requiring additional atomistic simulations during the prediction stage.



The proposed framework consists of two sequential stages: (i) generation of an MD dataset through low-energy nitrogen ion bombardment of FCC 316L stainless steel, and (ii) development and evaluation of machine learning regression models using incident ion energy and impact angle as input descriptors and MD-derived sputtering yield and penetration depth as target variables. By combining the physical accuracy of atomistic simulations with the computational efficiency of data-driven modeling, the integrated MD–ML framework provides a reliable surrogate model for predicting ion-induced surface modification over a wide range of irradiation conditions.

2.1 Molecular Dynamics Simulation Conditions

Molecular dynamics (MD) simulations were performed to generate the dataset used for machine learning prediction of sputtering yield and penetration depth during low-energy nitrogen ion bombardment of 316L stainless steel. The target material was modeled as a face-centered cubic (FCC) 316L stainless steel substrate with a (001) surface orientation. The simulation cell contained 1440 FCC lattice atoms generated using a $6 \times 6 \times 40$ replication scheme. Periodic boundary conditions were applied along the lateral directions, while a vacuum region of approximately 50 Å was introduced above the surface to provide sufficient space for ion interaction and sputtered atom emission.

The simulations were performed using an in-house C++ molecular dynamics code developed for ion–solid interaction studies. Prior to ion irradiation, the substrate was equilibrated at 300 K. Along the depth direction, the substrate was divided into three atomic regions: a fixed bottom layer, a thermostatted intermediate layer, and a fully dynamic region. The fixed layer prevented artificial translation of the substrate during ion impacts, while the thermostatted region acted as a heat sink to dissipate the thermal energy generated by collision cascades. The upper dynamic region allowed unrestricted atomic motion and contained the active collision zone where ion implantation, atomic displacement, and sputtering processes occurred. The equations of motion were integrated using the velocity-Verlet algorithm with a time step of 0.2 fs.

Nitrogen ions with incident energies of 50, 100, 200, and 300 eV were introduced at impact angles of 0°, 15°, 30°, 45°, 60°, and 75°. For each energy–angle combination, 3000 independent single-ion impact events were simulated using randomly distributed impact positions to obtain statistically reliable irradiation responses. To minimize artificial accumulation of irradiation damage, the atomic configuration was restored to the initial equilibrated state after every 1000 ion impacts.

The short-range interactions between incident nitrogen ions and substrate atoms were described using the Ziegler–Biersack–Littmark (ZBL) repulsive potential, whereas longer-range atomic interactions were represented using Morse and Lennard–Jones potentials. These interaction models were employed to describe the high-energy collision stage and subsequent atomic relaxation processes during ion bombardment.

The simulations provided atomic-scale information on ion trajectories, energy transfer, atomic displacement, sputtering events, and ion penetration behavior. The penetration depth of implanted nitrogen ions was defined as the maximum depth reached by nitrogen atoms along the z-direction, measured from the initial surface plane.

Sputtering yield and penetration depth represent two complementary responses of the ion–surface interaction process. While penetration depth describes subsurface ion transport and energy dissipation, sputtering yield reflects the fraction of transferred energy contributing to surface atom displacement and ejection. Therefore, these two parameters provide a comprehensive description of ion-induced surface modification, including subsurface implantation and surface erosion phenomena.

For each irradiation condition, sputtering yield and penetration depth were extracted as target variables for machine learning prediction. The sputtering yield was calculated as:

$$Y_s = \frac{N_{sputtered}}{N_{projectile}}, \quad 1$$

where ($N_{sputtered}$) represents the total number of atoms ejected from the 316L surface during 3000 ion impacts, and ($N_{projectile} = 3000$) represents the total number of incident nitrogen ions.

In addition to sputtering yield and penetration depth, the final atomic configurations of implanted nitrogen ions were analyzed to investigate implantation mechanisms. Nitrogen ions localized near Cr lattice sites with an N–Cr distance below 1 Å were classified as trapped states. Interstitial implantation was identified when nitrogen atoms remained in non-substitutional lattice positions with neighboring substrate atoms located within 2 Å after losing their kinetic energy through successive collision events [36].

2.2 Machine Learning Methodology

The machine learning task was formulated as a supervised regression problem, in which the objective was to predict sputtering yield and penetration depth directly from the irradiation conditions [3,4,15]. The input feature vector consisted of two experimentally controllable parameters:

$$X = [E_{incident}, \theta_{impact}], \quad 2$$

where ($E_{incident}$) is the incident ion energy (eV) and (θ_{impact}) is the impact angle (degree). These descriptors were selected because they directly represent the controllable irradiation conditions and enable prediction without requiring additional atomistic information.

Regression models were trained to learn the relationship between the irradiation parameters and the corresponding MD responses. Once trained, the models can rapidly predict sputtering yield and penetration depth for new irradiation conditions without performing additional molecular dynamics simulations, thereby significantly reducing the computational cost.

To evaluate the predictive capability of different machine learning approaches, three regression algorithms with increasing model complexity were investigated: Linear Regression (LR), Decision Tree Regression (DTR), and Random Forest Regression (RFR). Linear Regression was employed as a baseline model, whereas Decision Tree and Random Forest were used to capture nonlinear relationships between the irradiation parameters and the target responses. The performance of all models was assessed using standard regression metrics, including the coefficient of determination (R^2), mean absolute error (MAE), and root mean square error (RMSE).

A five-fold cross-validation strategy was employed to evaluate the generalization capability of the models. In each fold, approximately 80% of the samples were used for training and 20% for validation. Although 3000 independent ion impacts were simulated for each irradiation condition, the averaged sputtering yield and penetration depth values were used as representative outputs for

machine learning. Therefore, the final dataset consisted of 48 unique irradiation conditions obtained from four ion energies and six impact angles.

Regression Models

Three regression algorithms with different levels of complexity were employed to predict both sputtering yield and penetration depth from the molecular dynamics' dataset. Although the dataset consisted of only 48 averaged irradiation conditions, the framework demonstrates the feasibility of integrating molecular dynamics simulations with machine learning. Future work will employ larger datasets covering wider energy ranges and additional irradiation parameters.

Linear Regression (LR):

Linear Regression was employed as a baseline model to establish a linear relationship between the input features and the target variables. Owing to its simplicity and high interpretability, LR provides a useful benchmark for evaluating the benefits of more advanced nonlinear models.

Decision Tree Regression (DTR):

Decision Tree Regression was used to model nonlinear relationships by recursively partitioning the feature space into a series of decision rules. This approach can capture complex interactions between incident energy, impact angle, and the target responses without assuming linearity.

Random Forest Regression (RFR):

Random Forest Regression was selected as the primary predictive model because of its ability to accurately learn complex nonlinear relationships while reducing overfitting. By combining predictions from multiple decision trees trained on randomly sampled subsets of the data, the model provides improved prediction accuracy, robustness, and generalization.

ПОВТОР

The predictive performance of all models was evaluated using three standard regression metrics: the coefficient of determination (R^2), the root means square error (RMSE), and the mean absolute error (MAE). Comparing the three algorithms enables assessment of the trade-off between model simplicity, interpretability, and predictive accuracy, while identifying the most suitable regression approach for predicting ion-induced surface responses from molecular dynamics data.

The Random Forest model was constructed using 200 decision trees (estimators = 200) with a fixed random state of 42 to ensure reproducibility.

3. Results and Discussion

3.1 Dataset Characteristics and Sputtering Yield Analysis

The generated dataset revealed a clear dependence of sputtering yield on both incident ion energy and impact angle. Figure 2 presents the variation of sputtering yield as a function of incident ion energy for different impact angles. At low incident energies, particularly below 100 eV, sputtering

events were rarely observed, with sputtering yields below 0.10 for all impact angles. This is because the transferred kinetic energy from nitrogen ions to surface atoms was insufficient to overcome the surface binding energy of 316L stainless steel (approximately 4.5 eV). Under these conditions, most incident ions were either reflected from the surface or trapped in shallow interstitial sites near the surface region without causing significant atomic displacement.

With increasing ion energy from 100 to 300 eV, the probability of sputtering increased significantly. The sputtering yield at normal incidence (0°) increased from approximately 0.08 at 50 eV to 0.25 at 300 eV, representing a three-fold increase. This enhancement is attributed to the formation of more energetic collision cascades and enhanced momentum transfer between incident nitrogen ions and target atoms. The higher-energy impacts generated larger atomic displacements, increasing the likelihood of surface atom ejection. In addition, the influence of impact angle became more pronounced at elevated energies. Near-normal incidence conditions (0° – 15°) promoted deeper penetration and stronger collision cascades, whereas oblique impacts (60° – 75°) increased the probability of ion reflection and surface interaction, reducing the effective energy available for sputtering.

The angular dependence of sputtering yield exhibited a non-monotonic behavior. At intermediate impact angles (30° – 45°), the sputtering yield reached its maximum value across all energy levels. For example, at 300 eV, the sputtering yield increased from 0.25 at normal incidence to approximately 0.32 at 30° – 45° , before decreasing to 0.18 at 75° . This behavior reflects the competing effects of increased energy deposition near the surface at oblique angles and enhanced ion reflection at grazing angles. At 75° , a significant fraction of incident ions is reflected from the surface, reducing the effective energy deposition and consequently decreasing the sputtering yield.

The obtained dataset therefore represents the complex nonlinear relationship between incident energy, impact angle, and sputtering response, with the sputtering yield varying by approximately a factor of four across the investigated parameter space. These characteristics highlight the necessity of employing machine learning approaches capable of capturing nonlinear dependencies for accurate prediction of sputtering yield.

похибки!

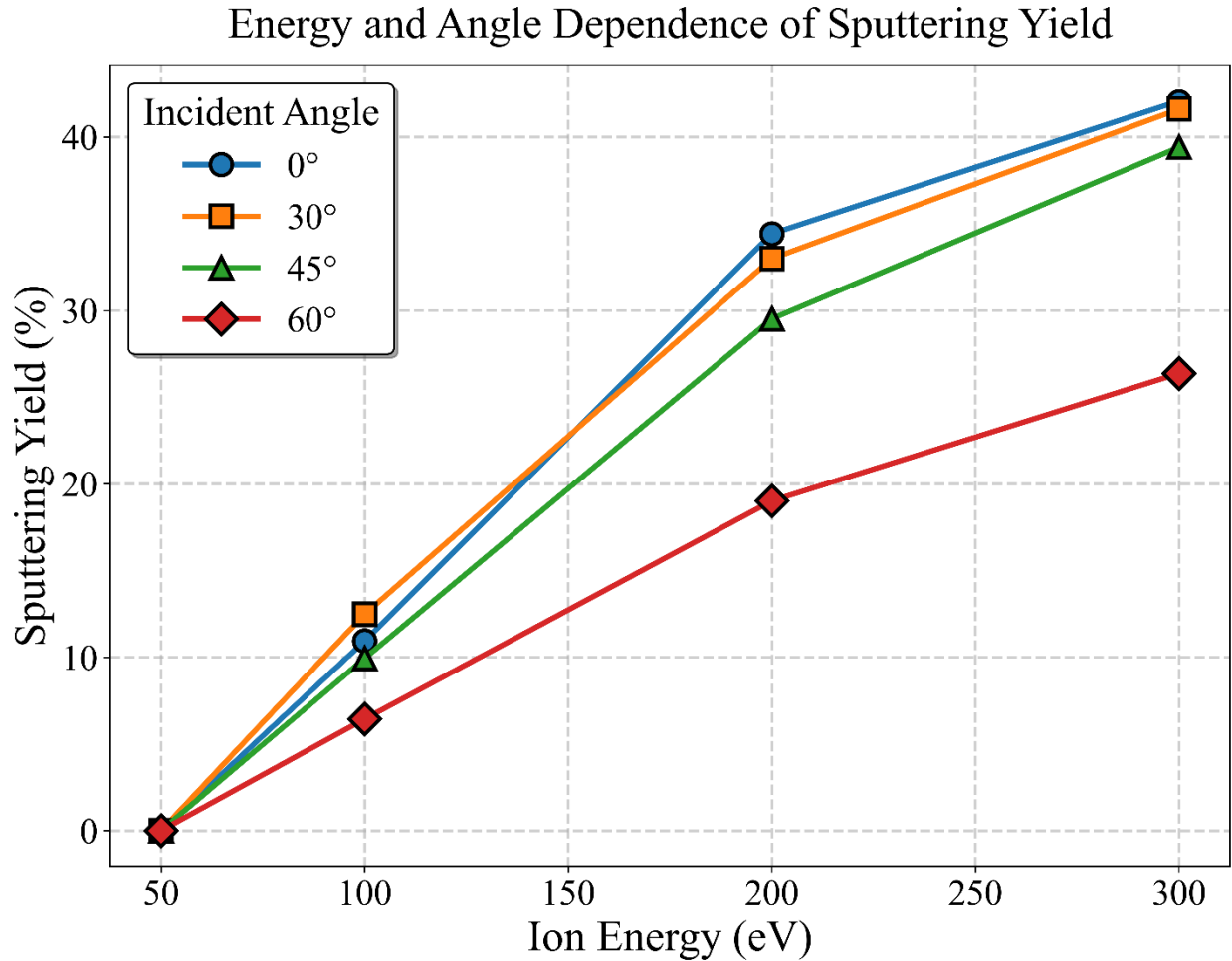


Figure 2. Energy and incidence-angle dependence of sputtering yield of nitrogen ions in 316L stainless steel obtained from molecular dynamics simulations.

Figure 2. Energy and incidence-angle dependence of sputtering yield for nitrogen ion bombardment of 316L stainless steel obtained from molecular dynamics simulations. The sputtering yield increases markedly with incident ion energy from 50 to 300 eV. Its angular dependence exhibits a non-monotonic behavior, with maximum values observed at intermediate incidence angles (30°–45°). At 50 eV, sputtering is negligible for all investigated angles because the transferred kinetic energy is insufficient to overcome the surface binding energy (~ 4.5 eV). At higher energies, enhanced collision cascades and more efficient near-surface energy deposition at intermediate angles promote surface atom ejection, whereas sputtering decreases at grazing incidence due to increased ion reflection and reduced normal energy transfer.

Figure 3. Energy and incidence-angle dependence of nitrogen ion penetration depth in 316L stainless steel obtained from molecular dynamics simulations. The penetration depth increases with incident ion energy because higher kinetic energy promotes deeper ion implantation through enhanced momentum transfer and collision cascades. Increasing the incidence angle reduces the effective normal energy component ($E_{\perp} = E \cos^2\theta$), resulting in shallower penetration due to increased lateral scattering and ion reflection.

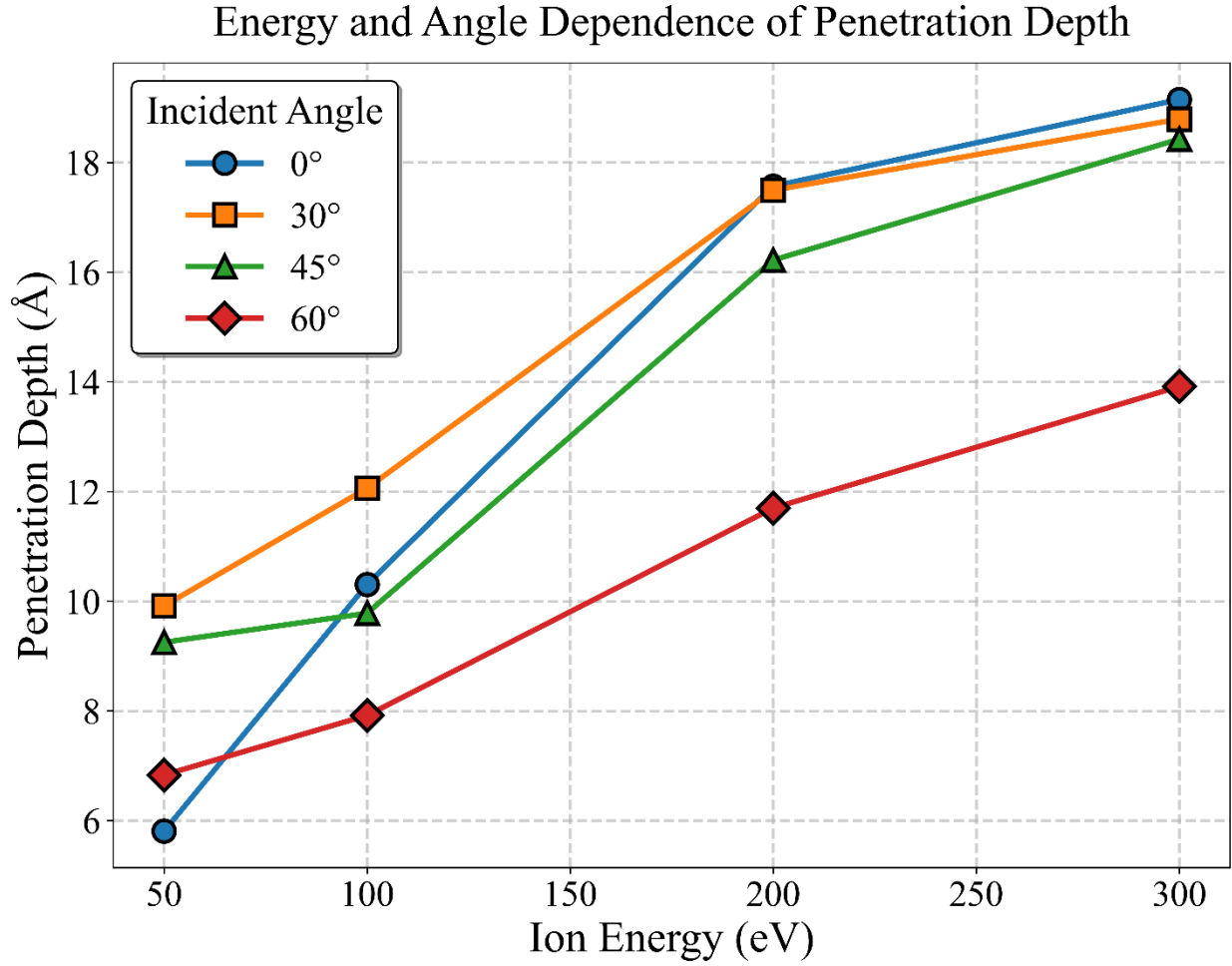


Figure 3. Energy and incidence-angle dependence of penetration depth of nitrogen ions in 316L stainless steel obtained from molecular dynamics simulations.

At low energies (50–100 eV), the penetration depth remained limited because the transferred kinetic energy was insufficient to generate extensive collision cascades. With increasing energy up to 300 eV, the penetration depth increased significantly, particularly for normal and near-normal incidence conditions. The reduction observed at larger incidence angles is attributed to the decrease in the perpendicular energy component ($E_{\perp} = E \cos^2 \theta$) and increased probability of surface scattering.

Feature Importance for Penetration Depth

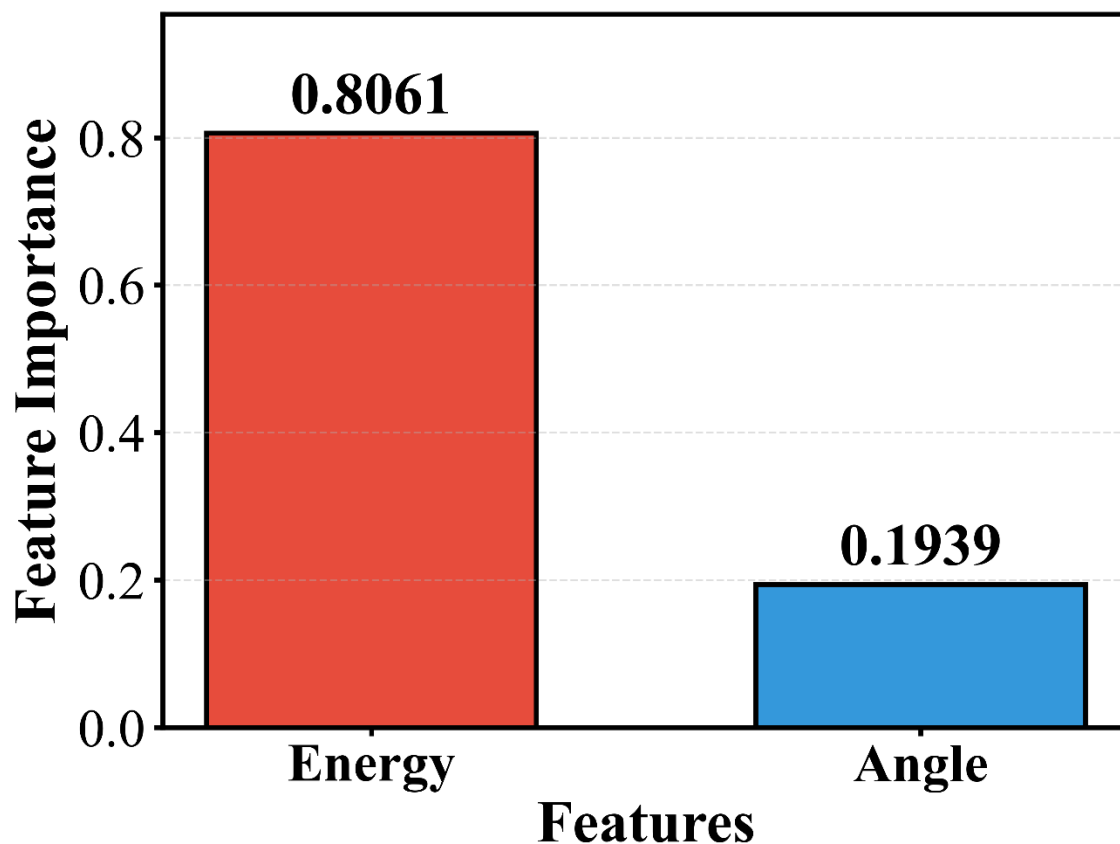


Figure 4. Feature importance analysis of the Random Forest regression model for predicting penetration depth (Depth_A) in 316L stainless steel under nitrogen ion bombardment. The bar chart displays the relative contribution of two input features: Energy (red) and Angle (blue). Energy exhibits significantly higher importance (0.7812) compared to Angle (0.2188), indicating that beam energy is the dominant factor controlling the penetration depth during ion bombardment. This finding is physically consistent with MD simulations, where higher ion energies promote deeper penetration through enhanced momentum transfer and more extensive collision cascades.

Feature Importance for Sputtering Yield

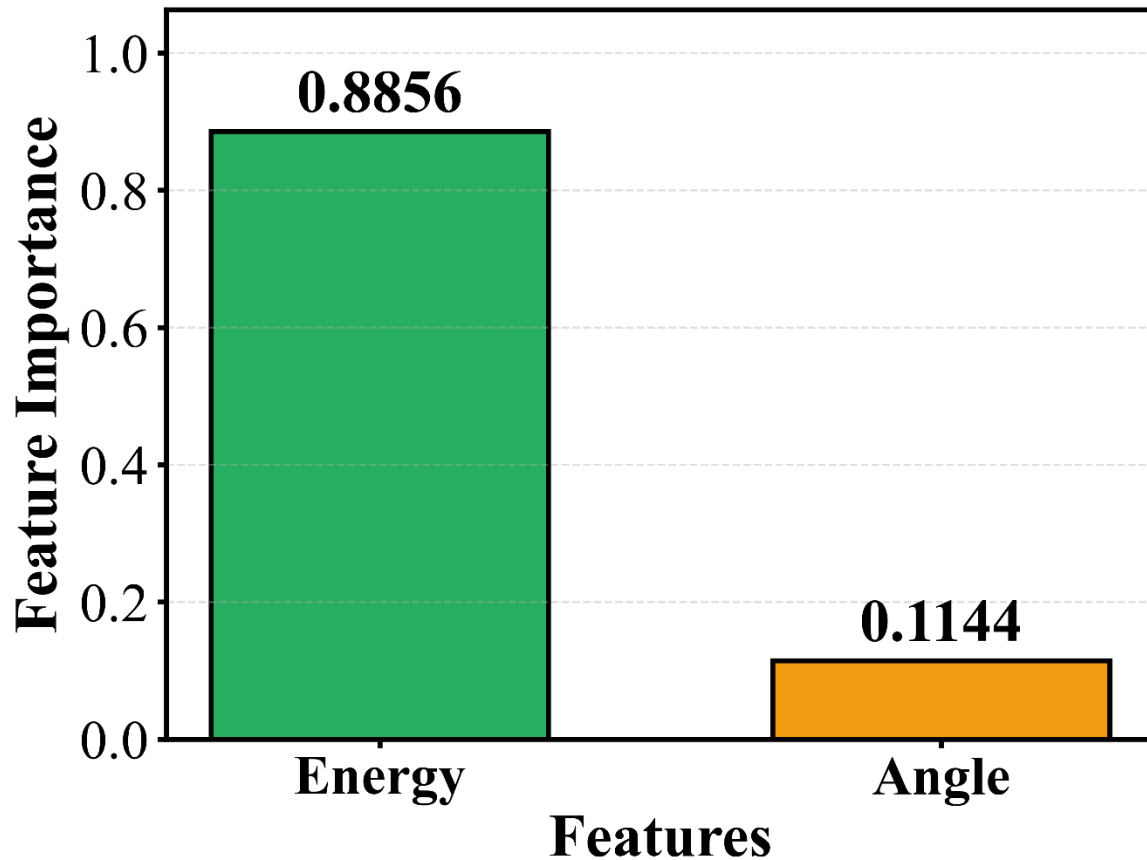


Figure 5. Feature importance analysis of the Random Forest regression model for predicting sputtering yield (Sputtered_%) in 316L stainless steel under nitrogen ion irradiation. The bar chart illustrates the relative contribution of two input features: Energy (green) and Angle (orange). The analysis reveals that Energy has a substantially higher importance (0.8856) compared to Angle (0.1144), suggesting that the incident ion energy plays a dominant role in determining the sputtering yield. This finding is physically reasonable since higher ion energies lead to more efficient momentum transfer and atomic displacement cascades, resulting in increased material ejection from the surface. Although Angle has a lower impact, its influence remains non-negligible, as it affects the energy deposition profile and collision cascade geometry.

Random Forest: Penetration Depth Prediction

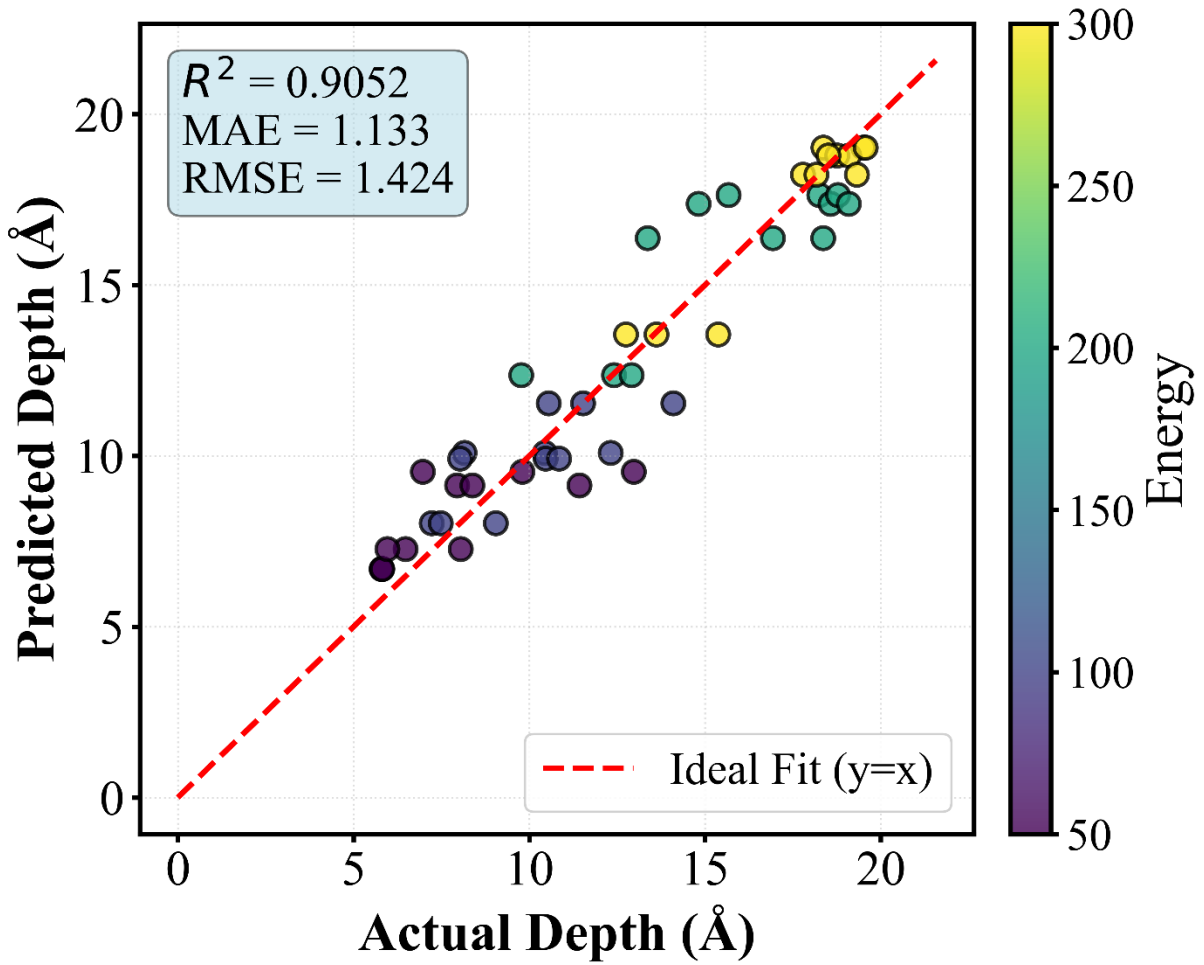


Figure 6. Performance evaluation of the Random Forest regression model for predicting penetration depth (Depth_A) in SS316 stainless steel during nitrogen ion bombardment. The scatter plot compares predicted versus actual depth values, with points color-coded by beam energy (ranging from low to high). The dashed diagonal line represents the ideal prediction ($y=x$). The model achieves excellent predictive performance with $R^2 = 74.7\%$, MAE = 1.830 Å, and RMSE = 2.196 Å, demonstrating that 74.7% of the variance in penetration depth is explained by the input features. The close alignment of data points along the diagonal line confirms the model's high accuracy and reliability. This data-driven approach provides a rapid and cost-effective tool for predicting penetration depth across various nitrogen ion bombardment conditions, significantly reducing the need for extensive experimental trials.

Random Forest: Sputtering Yield Prediction

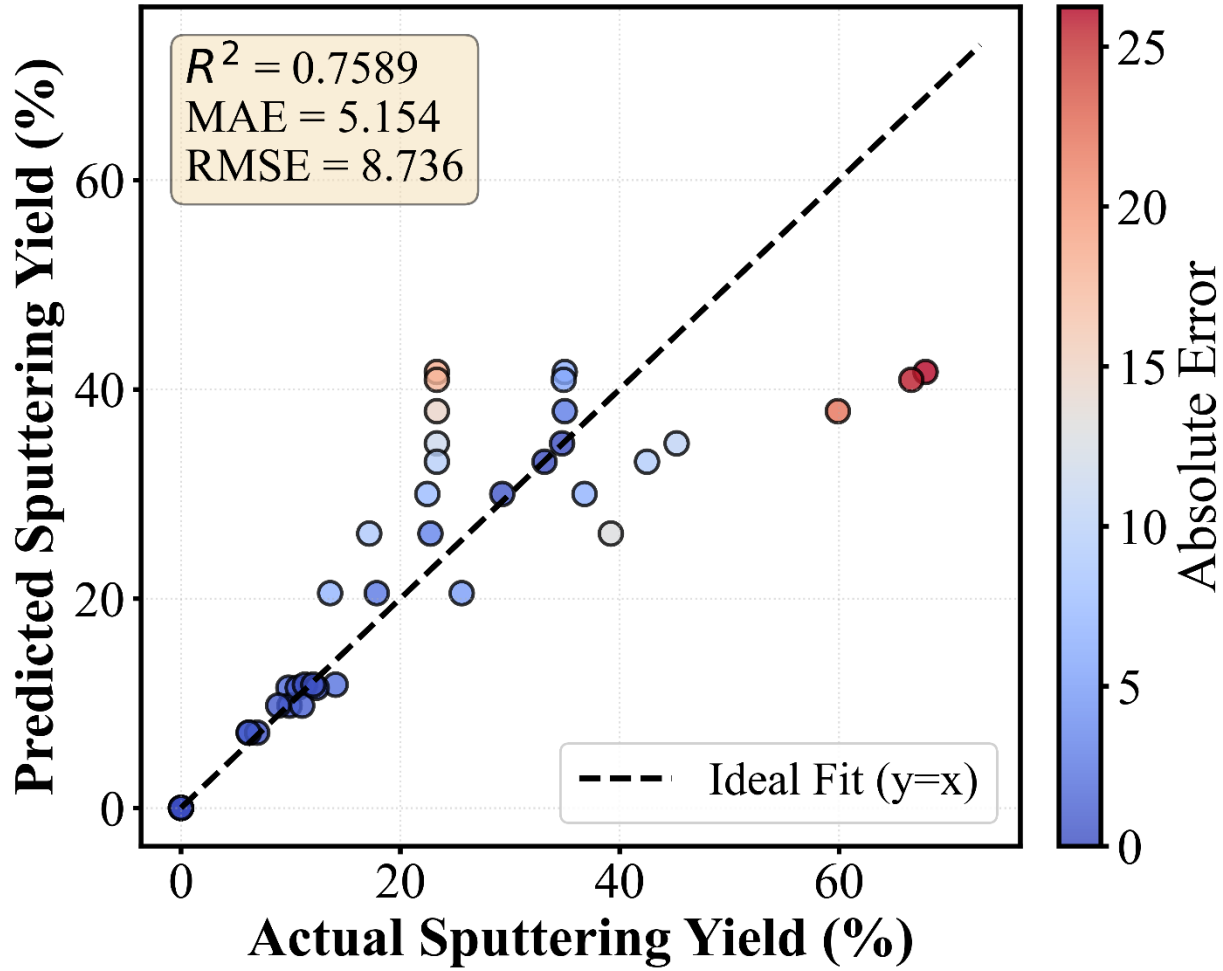


Figure 7. Performance of the Random Forest regression model for predicting sputtering yield in 316L stainless steel under low-energy nitrogen ion bombardment. The scatter plot compares the predicted and actual sputtering yield values, while the dashed diagonal line represents the ideal prediction ($y = x$). The model achieved an R^2 value of 0.572, an RMSE of 10.746, and an MAE of 6.763. Although the prediction accuracy is lower than that obtained for penetration depth, the Random Forest model captures the overall nonlinear relationship between irradiation conditions and sputtering yield, providing a useful surrogate model for this inherently stochastic process.

Figures 6 and 7 compare the predicted and actual values for penetration depth and sputtering yield obtained using the Random Forest model. Figure 6 demonstrates excellent predictive capability for penetration depth, whereas Figure 7 indicates moderate prediction accuracy for sputtering yield owing to the intrinsically stochastic nature of sputtering.

3.2 Performance Comparison of Regression Models

The predictive performance of the developed regression models was evaluated using five-fold cross-validation to assess their generalization capability and reliability. Three regression algorithms, including Linear Regression (LR), Decision Tree Regression (DTR), and Random Forest Regression (RFR), were investigated for predicting sputtering yield and interaction depth. The models were quantitatively compared using the coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE).

Linear Regression was first applied as a baseline model to examine the extent to which the relationship between ion bombardment parameters and target responses could be described by a linear approximation. For sputtering yield prediction, LR achieved an R^2 value of 0.586, indicating that a considerable part of the variation in sputtering behavior can be correlated with incident energy and impact angle. However, the remaining prediction error reflects the nonlinear characteristics of collision cascades, atomic displacement processes, and angular effects during ion irradiation.

Decision Tree Regression showed lower performance for sputtering yield prediction, with an R^2 value of 0.282. Although decision trees are capable of modeling nonlinear relationships, the relatively limited size and complexity of the dataset may increase sensitivity to individual data variations and reduce generalization capability.

має бути розмірне

Random Forest Regression provided comparable prediction performance for sputtering yield, achieving an R^2 value of 0.572 and the lowest MAE value (6.763), whereas Linear Regression achieved the highest R^2 (0.586) and the lowest RMSE (10.502). This suggests that although the linear model explained slightly more of the overall variance, the ensemble model produced smaller average prediction errors. The results indicate that sputtering yield prediction remains challenging because of the highly nonlinear and stochastic nature of ion-induced atomic processes.

Table 1. Performance comparison of regression models for sputtering yield prediction

Model	R^2	RMSE	MAE
Linear Regression	0.586	10.502	7.305
Decision Tree Regression	0.282	13.500	8.348
Random Forest Regression	0.572	10.746	6.763

For interaction depth prediction, Random Forest Regression demonstrated superior performance compared with LR and DTR. The model achieved the highest R^2 value (0.747) and the lowest RMSE (2.196 Å) and MAE (1.830 Å), confirming its capability to capture the nonlinear dependence between ion energy, impact angle, and penetration behavior.

Table 2. Performance comparison of regression models for interaction depth prediction

Model	R ²	RMSE (Å)	MAE (Å)
Linear Regression	0.683	2.469	1.893
Decision Tree Regression	0.667	2.521	2.163
Random Forest Regression	0.747	2.196	1.830

The superior performance of Random Forest for depth prediction can be attributed to its ensemble structure, which combines multiple decision trees and effectively captures complex nonlinear interactions between input variables. These findings demonstrate that machine learning models can provide efficient surrogate predictions for molecular dynamics simulations, reducing computational requirements for extensive parameter exploration.

3.3 Prediction Performance of the Random Forest Model

Figures 6 and 7 illustrate the predictive capability of the Random Forest model for penetration depth and sputtering yield, respectively. The model predicts penetration depth with relatively high accuracy ($R^2 = 0.747$), whereas the prediction of sputtering yield is less accurate ($R^2 = 0.572$). This difference is consistent with the underlying physical mechanisms, as penetration depth varies more systematically with incident energy and impact angle, while sputtering is governed by highly stochastic collision cascades and surface atom ejection events. These results indicate that Random Forest is particularly suitable for predicting penetration behavior, while sputtering yield remains more challenging because of its inherent variability.

Five-fold cross-validation was employed. In each fold, approximately 80% of the data were used for training and the remaining 20% for testing.

3.4 Discussion of Feature Influence

The feature importance analysis obtained from the Random Forest model confirms that incident ion energy is the dominant predictor for both penetration depth and sputtering yield. Energy contributes substantially more than impact angle, which is consistent with the molecular dynamic's simulations showing that higher ion energies generate stronger collision cascades, greater momentum transfer, and deeper ion penetration. Although the impact angle has a smaller overall contribution, it remains an important factor because it modifies the normal component of the incident energy and influences both energy deposition near the surface and the probability of atom ejection.

3.5 Limitations of the Present Study

The present study is based on a molecular dynamics-generated dataset covering a limited range of ion energies and impact angles. Consequently, the predictive capability of the developed model is restricted to the investigated irradiation conditions. Expanding the dataset by considering additional energies, impact angles, ion species, and target materials would improve the generalization capability of the machine learning model. Future studies may also investigate deep

learning approaches and physics-informed machine learning models using larger simulation datasets.

4. Conclusions

In this proof-of-concept study, an MD-informed machine learning framework was developed to predict sputtering yield and penetration depth during low-energy nitrogen ion bombardment of 316L stainless steel. The MD simulations revealed that sputtering yield is primarily governed by near-surface energy deposition, increasing with ion energy as collision cascades become more energetic and momentum transfer to surface atoms intensifies. The angular dependence of sputtering arises from variations in the normal energy component ($E_{\perp} = E \cos^2\theta$), which controls the depth of energy deposition and the probability of surface atom ejection. At intermediate angles (30° – 45°), the balance between sufficient normal energy for subsurface cascades and reduced ion reflection maximizes sputtering, whereas at grazing angles, enhanced reflection reduces effective energy transfer.

Penetration depth, in contrast, is determined by subsurface momentum transport and the stopping power experienced by incident ions. Higher ion energies promote deeper implantation through more energetic recoil cascades, while oblique impact angles reduce the normal momentum component, leading to shallower penetration and increased lateral scattering. This fundamental difference in underlying physics—surface-dominated energy deposition versus subsurface momentum transport—explains why sputtering yield and penetration depth exhibit distinct dependencies on irradiation parameters.

Consistent with these physical mechanisms, Linear Regression captured the sputtering yield adequately ($R^2 = 0.586$), suggesting that the averaged sputtering response follows a roughly linear trend with energy and angle within the investigated range. In contrast, Random Forest Regression provided superior performance for penetration depth ($R^2 = 0.747$, RMSE = 2.196 Å, MAE = 1.830 Å), effectively capturing the nonlinear coupling between ion energy, impact angle, and subsurface ion transport. This distinction highlights that the choice of ML model should be guided by the underlying physics of the target response.

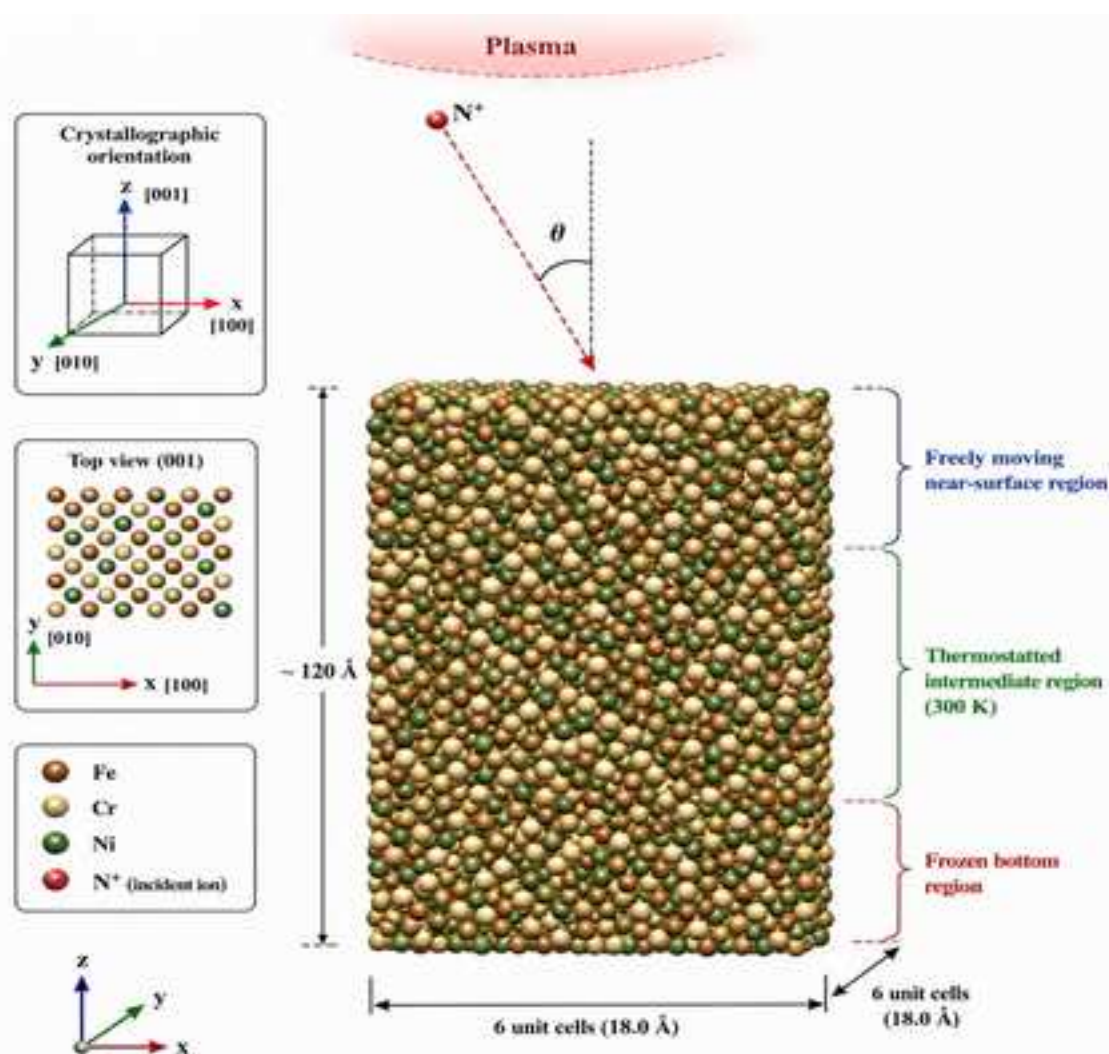
The MD–ML framework offers a computationally efficient surrogate modeling approach, reducing the need for extensive atomistic simulations while preserving physical interpretability through MD-generated training data. Although limited by the current dataset (48 unique irradiation conditions), this study establishes a feasible pathway for rapid screening of ion implantation parameters. The methodology is readily extendable to larger datasets, broader energy ranges, multiple ion species, and diverse target materials. Future work will focus on expanding the simulation database and incorporating physics-informed constraints or deep learning architectures to further enhance predictive accuracy, robustness, and generalization across wider processing windows.

Reference

- [1] Eckstein, W., *Computer Simulation of Ion-Solid Interactions*, Springer, Berlin, 1991.
- [2] Behrisch, R. (Ed.), Eckstein, W., *Sputtering by Particle Bombardment: Experiments and Computer Calculations from Threshold to MeV Energies*, Springer, Berlin, 2007.
- [3] Morgan, D., Jacobs, R., Opportunities and challenges for machine learning in materials science, *Annual Review of Materials Research*, **50** (2020) 71–103. <https://doi.org/10.1146/annurev-matsci-070218-010015>
- [4] Deringer, V.L., Caro, M.A., Csányi, G., Machine learning interatomic potentials as emerging tools for materials science, *Journal of Chemical Physics*, **153** (2020) 170901. <https://doi.org/10.1063/5.0017000>
- [5] Kino, H., Ikuse, K., Dam, H.C., Hamaguchi, S., Characterization of descriptors in machine learning for data-based sputtering yield prediction, *Physics of Plasmas*, **28** (2021) 013504. <https://doi.org/10.1063/5.0006816>
- [6] Smith, R., *Atomic and Ion Collisions in Solids and at Surfaces*, Cambridge University Press, Cambridge, 1997.
- [7] Sigmund, P., Theory of sputtering. I. Sputtering yield of amorphous and polycrystalline targets, *Physical Review*, **184** (1969) 383–416.
- [8] Nordlund, K., et al., Molecular dynamics simulation of ion ranges in the 1–100 keV energy range, *Nuclear Instruments and Methods in Physics Research Section B*, **203** (2003) 20–24.
- [9] Ziegler, J.F., Biersack, J.P., Littmark, U., *The Stopping and Range of Ions in Matter (SRIM)*, Pergamon Press, New York, 1985.
- [10] Nordlund, K., et al., Primary radiation damage: A review of mechanisms and modeling, *Nuclear Instruments and Methods in Physics Research Section B*, **406** (2018) 1–18.
- [11] Williamson, D.L., Davis, J.R., et al., Nitrogen implantation of stainless steel, *Materials Science and Engineering*, **69** (1985) 3–10.
- [12] Picard, S., Schneider, M., et al., Nitrogen ion implantation of 316L stainless steel, *Surface and Coatings Technology*, **203** (2009) 2633–2639.
- [13] Garvie, R.C., et al., Corrosion behavior of nitrogen-implanted 316L stainless steel, *Corrosion Science*, **70** (2013) 154–162.

- [14] Wang, Z., et al., Machine learning for materials science, *Nature Reviews Materials*, **4** (2019) 412–427.
- [15] Butler, K.T., Davies, D.W., Cartwright, H., Isayev, O., Walsh, A., Machine learning for molecular and materials science, *Nature*, **559** (2018) 547–555.
- [16] Kauwe, S.K., Graser, J., Vazquez-Mayagoitia, A., Sparks, T.D., Machine learning for sputtering yield prediction, *Journal of Applied Physics*, **128** (2020) 175901.
- [17] Davis, J.R., *Stainless Steels*, ASM International, Materials Park, OH, 1994.
- [18] Lo, K.H., Shek, C.H., Lai, J.K.L., Recent developments in stainless steels, *Materials Science and Engineering: R*, **65** (2009) 39–104.
- [19] Pethő, L., et al., Nitrogen ion implantation of stainless steel: A review, *Surface and Coatings Technology*, **365** (2019) 15–23.
- [20] Was, G.S., *Fundamentals of Radiation Materials Science: Metals and Alloys*, Springer, Berlin, 2007.
- [21] Sigmund, P., Sputtering by ion bombardment: Theoretical concepts, in: *Topics in Applied Physics*, Vol. 47, Springer, 1981, pp. 9–71.
- [22] Allen, M.P., Tildesley, D.J., *Computer Simulation of Liquids*, 2nd ed., Oxford University Press, Oxford, 2017.
- [23] Frenkel, D., Smit, B., *Understanding Molecular Simulation: From Algorithms to Applications*, 2nd ed., Academic Press, San Diego, 2002.
- [24] Rafii-Tabar, H., Modelling the nano-scale phenomena in condensed matter physics via computer-based numerical simulations, *Physics Reports*, **325** (2000) 239–310.
- [25] Schmidt, J., Marques, M.R.G., Botti, S., Marques, M.A.L., Recent advances and applications of machine learning in solid-state materials science, *npj Computational Materials*, **5** (2019) 83.
- [26] Ramprasad, R., Batra, R., Pilania, G., Mannodi-Kanakkithodi, A., Kim, C., Machine learning in materials informatics: Recent applications and prospects, *npj Computational Materials*, **3** (2017) 54.
- [27] Yang, L., et al., Machine learning prediction of sputtering yields based on atomic-scale descriptors, *Computational Materials Science*, **203** (2022) 111102.
- [28] Hobler, G., Betz, G., On the useful range of sputtering yields, *Nuclear Instruments and Methods in Physics Research Section B*, **180** (2001) 203–208.

- [29] Averback, R.S., Diaz de la Rubia, T., Displacement damage in irradiated metals and semiconductors, *Solid State Physics*, **51** (1998) 281–402.
- [30] Hastie, T., Tibshirani, R., Friedman, J., *The Elements of Statistical Learning*, 2nd ed., Springer, New York, 2009.
- [31] James, G., Witten, D., Hastie, T., Tibshirani, R., *An Introduction to Statistical Learning*, Springer, New York, 2013.
- [32] Kalinin, S.V., et al., Machine learning in materials science: Recent progress and emerging challenges, *MRS Bulletin*, **46** (2021) 1041–1049.
- [33] Taghinejad, J., Niknam, A.R., Rastkar, A.R., Ghomi, H., Time-resolved evolution of collisional transient sheath in plasma source ion implantation, *Physica Scripta*, **96** (12) (2021) 125623.
- [34] Taghinejad, J., Niknam, A.R., Ghahyazi, M.R., Moradkhani, A., Molecular dynamics simulations of Ar⁺ bombardment of Fe (001), (101), and (111) surfaces: Study of threshold energy and angular characteristics of sputtering, *Vacuum*, **215** (2023) 112312.
- [35] Taghinejad, J., Niknam, A.R., Effects of collisionless and collisional plasma characteristics on the transport mechanism of nitrogen ions in nitrided austenitic stainless steel, *Plasma Chemistry and Plasma Processing*, **43** (4) (2023) 921–931.
- [36] Taghinejad, J., Vejdanihemmat, M., Niknam, A.R., Numerical analysis of nitrogen adsorption and penetration depth in plasma-nitrided austenitic stainless steel 316: A crystallographic orientation study, *AIP Advances*, **15** (4) (2025).



Ion Energy (50–300 eV)

+

Impact Angle (0–75°)

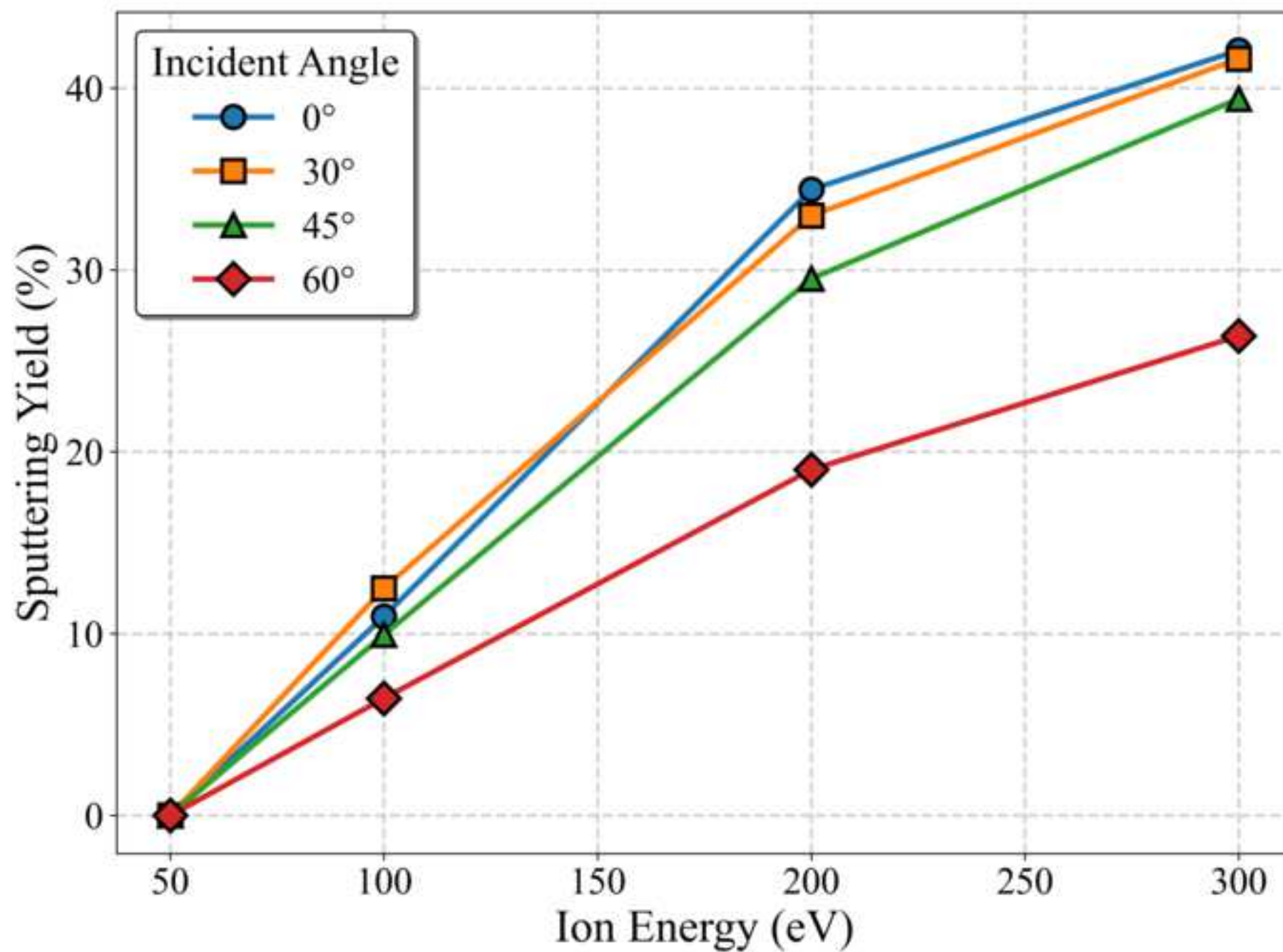
Molecular Dynamics
(3000 impacts/condition)

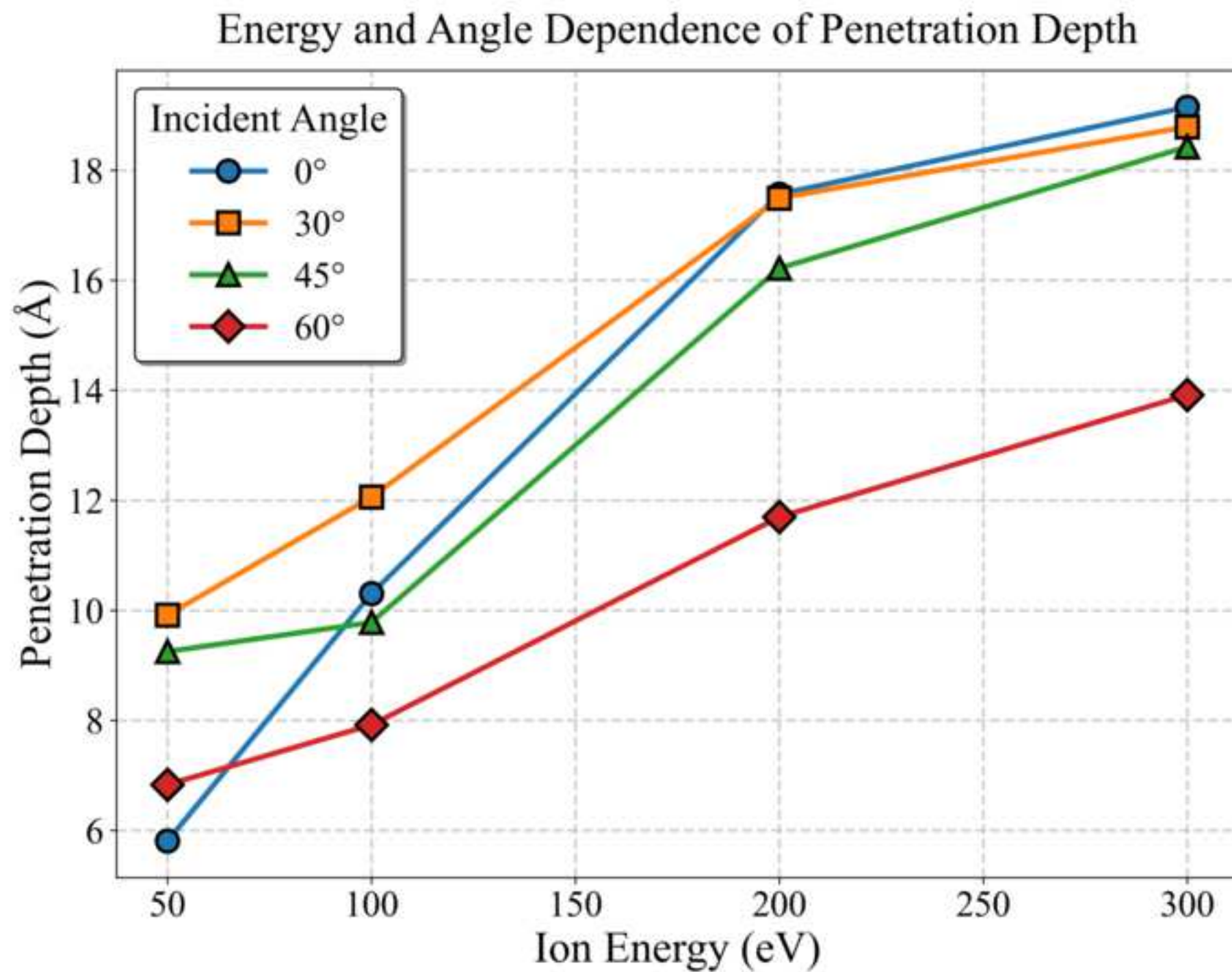
Sputtering Yield
Penetration Depth

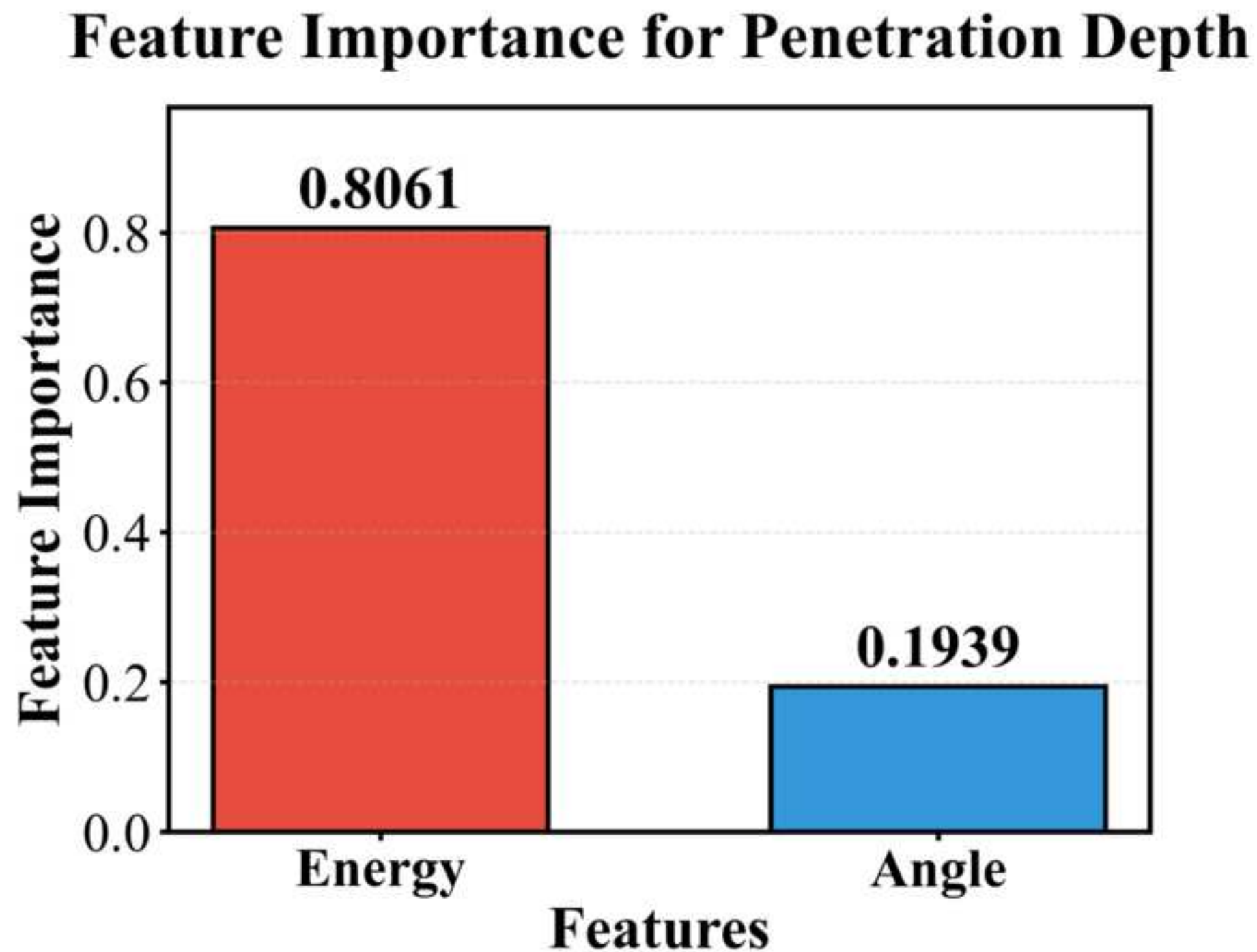
Machine Learning
LR | DTR | RFR

Prediction

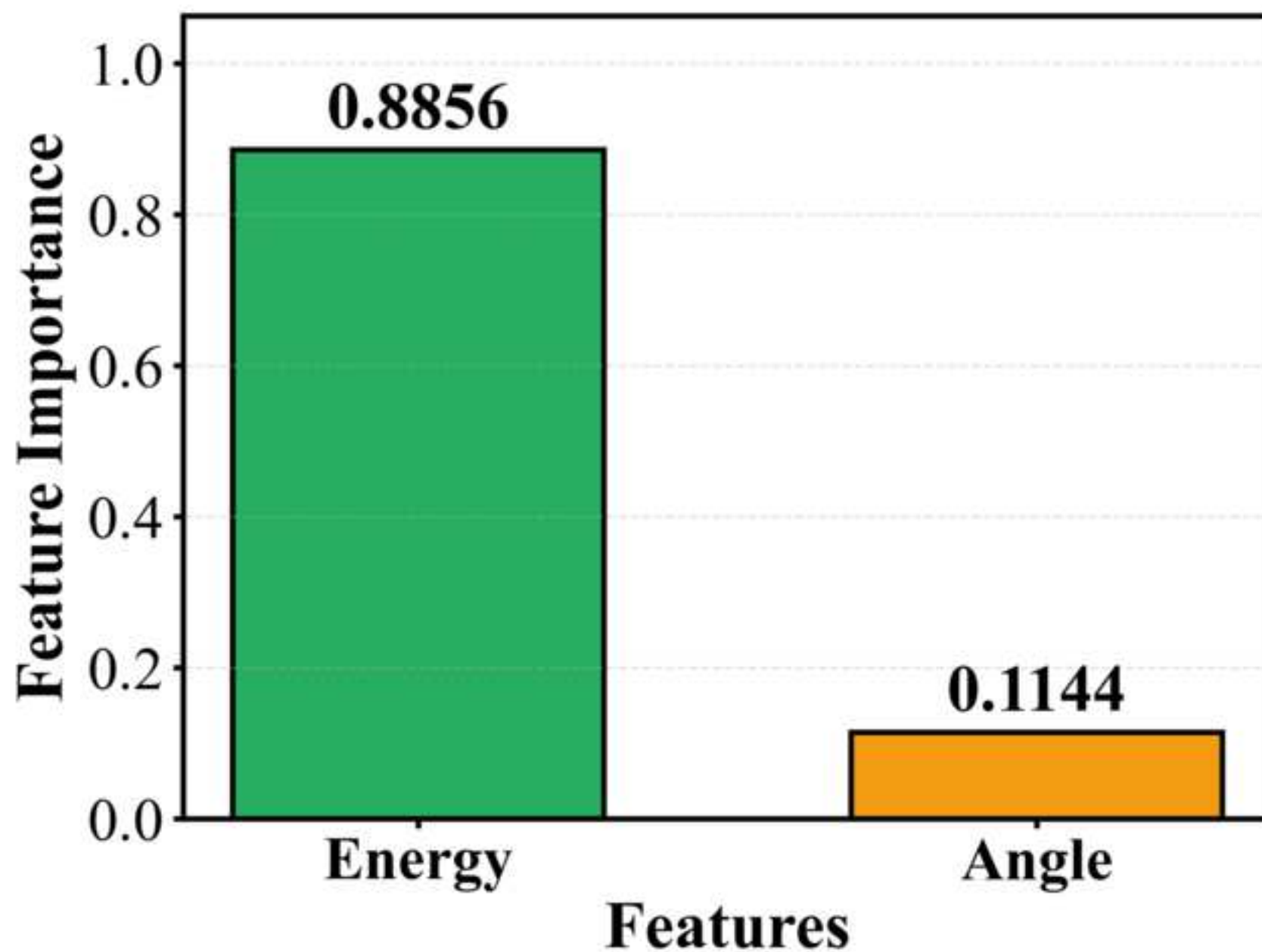
Energy and Angle Dependence of Sputtering Yield



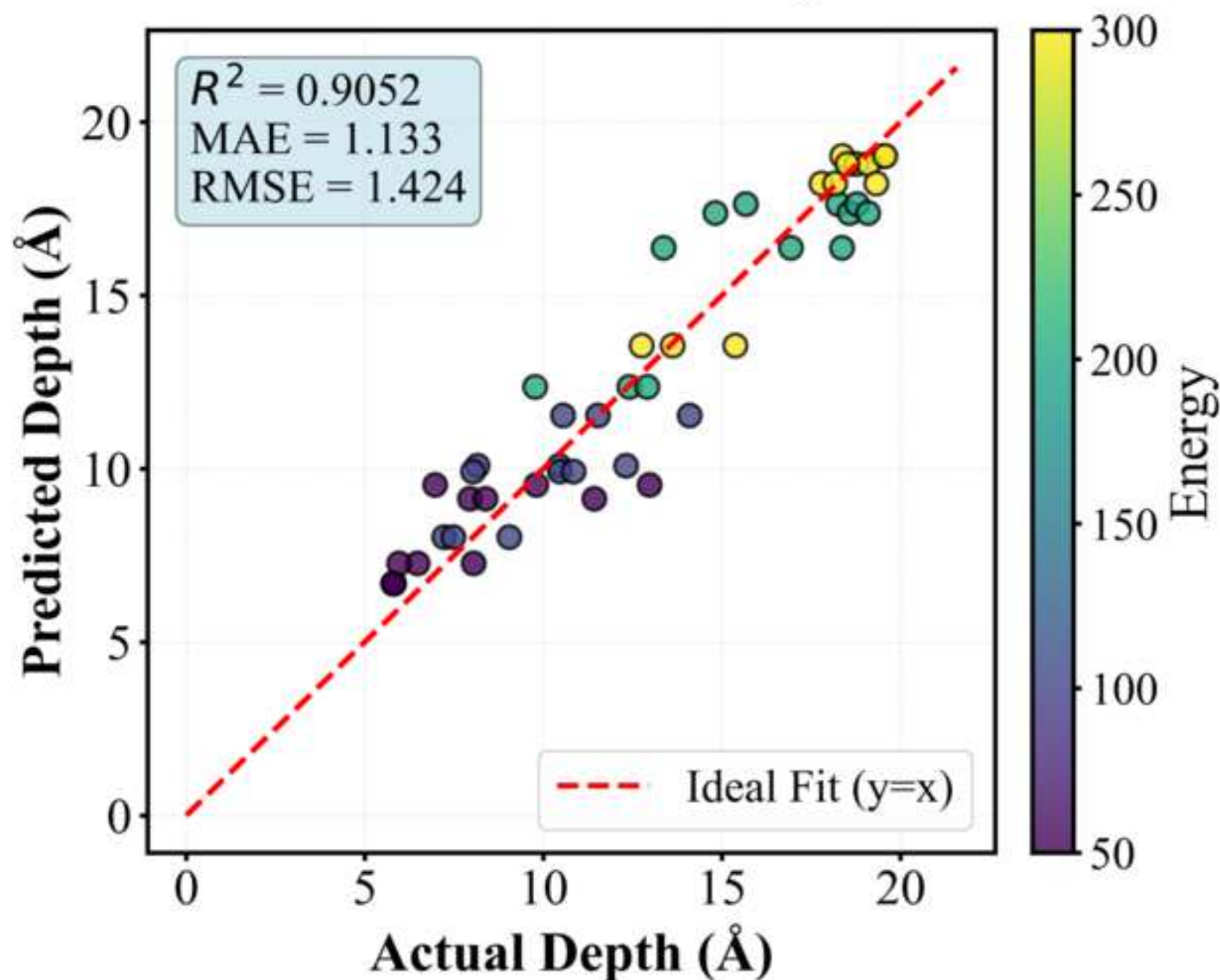




Feature Importance for Sputtering Yield



Random Forest: Penetration Depth Prediction



Random Forest: Sputtering Yield Prediction

